DIGITAL SIGNAL PROCESSING MIDTERMS

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CLASSIFICATION OF
DISCRETE TIME SYSTEM

Static:

- A discrete-time system is called **static** or **memoryless** if its output at any instant n depends at most on the input sample at the same time, but not on past or future samples of the input.
- There is no need to store any of the past inputs or outputs in order to compute the present output.
- The system described by the following input-output equations

$$y[n] = ax[n]$$
$$y[n] = nx[n] + bx^3[n]$$

are both static or memoryless.

Dynamic System:

- System that have a memory.
- If the output of the system at time n is completely determined by the input samples in the interval from n-N to n(N≥0), the system is said to have memory of duration N.

- If 0<N<∞, the system is said to have finite memory
- If N=∞, the system is said to have a infinite memory.
- The system described by the following input-output equations

$$y[n] = x[n] + 3x[n-1] \qquad y[n] = \sum_{k=0}^{n} x[n-k]$$
$$y[n] = \sum_{k=0}^{\infty} x[n-k]$$

are dynamic systems or system with memory.

Time Invariant:

- If its input-output characteristic do not change with time.
- **Definition.** A relaxed system is time-invariant or shift invariant if and only if

$$x[n] \xrightarrow{\mathcal{T}} y[n]$$

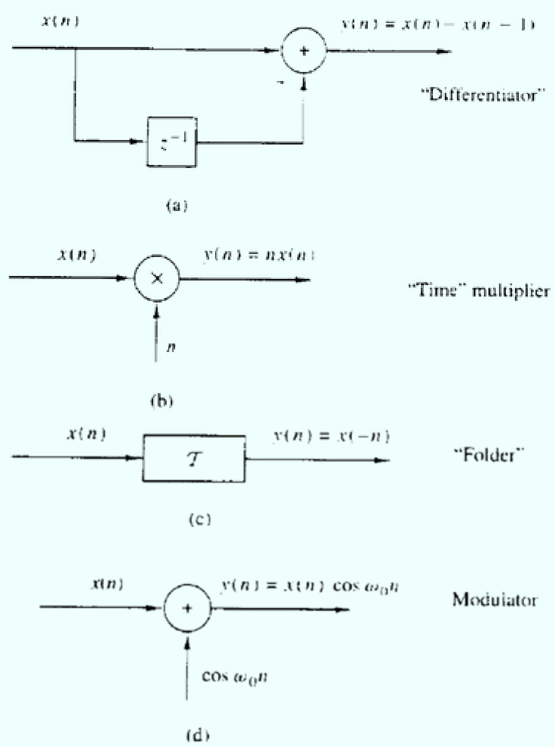
- implies that

$$x[n-k] \xrightarrow{\mathcal{T}} y[n-k]$$

- for every input signal x[n] and every time shift k.

How to determine if the system is time variant or time invariant ? To determine if any given system is time invariant, we need to perform the test specified by the preceding definition.
1.We excite the system with an arbitrary input sequence x[n], which produces an output denoted as y[n].
2.We delay the input sequence by some amount k and recomputed the output. In general, we can write the output as
y[n,k]=T[x[n-k]].
3.If the output y[n,k]=y[n-k], for all possible values of k, the system is time- invariant.
If the output y[n,k]≠y[n-k], even for one value of k, the system is time variant.

Example: Determine if the system shown below are time invariant or time variant.



CLASSIFICATION OF
DISCRETE TIME SYSTEM

Linear:

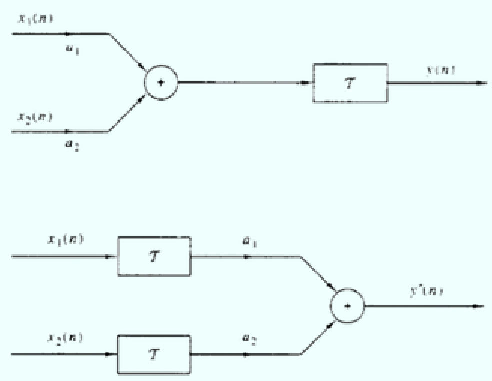
- It is one that satisfies the *superposition principle.*
- A relaxed ,linear system with zero input produces a zero output
- **Superposition Principle** –requires that the response of the system to a weighted sum of signals be equal to the corresponding weighted sum of the responses (outputs) of the system to each of the individual input signals.
- **Definition.** A relaxed T system is linear if and only if

$$\mathcal{T}[a_1x_1[n] + a_2x_2[n]] = a_1\mathcal{T}[x_1[n]] + a_2\mathcal{T}[x_2[n]]$$

- **f**or any arbitrary input sequence x_1 [n] and x_2 [n], and any arbitrary constant a_1 and a_2

Non-linear system:

- If a relaxed system does not satisfy the superposition principle as given by the definition above
- If a system produces a nonzero output with a zero input, the system may be either nonrelaxed or nonlinear
- T is linear if and only if y[n]=y^′ [n]



Casual vs. Non-casual

- **Definition.** A system is said to be causal if the output of the system at any time n depends only on present and past inputs, but does not depend on future inputs.
- If a system does not satisfy this definition, is called **noncausal**

- Stability is an important property that must be considered in any practical application of a system . Unstable systems usually exhibit erratic and extreme behavior and cause overflow in any practical implementation
- Definition. An arbitrary relaxed system is said to be bounded input-bounded output (BIBO) stable if and only if every bounded input produces a bounded output.

DISCRETE FOURIER TRANSFORM

- mathematical algorithm that can be computed efficiently on digital computers and other DSP Boards.
- an extension of CTFT for the time limited sequences with an additional restriction that the frequency is discretized at a finite set of values.

DFT Analysis Equation

$$x[k] = \sum_{n=0}^{N-1} x[n]e^{-j2\pi\frac{kn}{N}}$$

DFT Synthesis Equation: (inverse DFT)

$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} x[k]e^{j2\pi\frac{kn}{N}}$$

Calculate the four-point DFT of aperiodic sequence x[k] of length N = 4 which is defined by:

$$x[n] = \{2,3,-1,1\}$$

$$x[k] = \sum_{n=0}^{3} x\,[n]e^{-j_{2\Pi}\frac{kn}{4}}$$

$$x[0]e^{-j2\Pi}\frac{(0)(0)}{4} + x[1]e^{-j2\Pi}\frac{(0)(1)}{4} + x[2]e^{-j2\Pi}\frac{(0)(2)}{4} x[3]e^{-j2\Pi}\frac{(0)(3)}{4}$$

$$x[0] + x[1] + x[2] + x[3]$$

$$2 + 3 + (-1) + 1$$

$$5$$

DISCRETE FOURIER TRANSFORM (CONT.)

$$x[1] = \sum_{n=0}^{3} x\,[n]e^{-j2\Pi}\frac{(1)(n)}{4}$$

$$x[0]e^{-j2\Pi}\frac{(1)(0)}{4} + x[1]^{-j2\Pi}\frac{(1)(1)}{4} + x[2]^{-j2\Pi}\frac{(1)(2)}{4} + x[3]^{-j_{2\Pi}}\frac{(1)(3)}{4}$$

$$x[0] + x[1]e^{-j\frac{\Pi}{2}} + x[2]e^{-j\Pi} + x[3]e^{-j\frac{3\Pi}{2}}$$

$$2 + (3)\left(\cos\frac{\Pi}{2} - \mathrm{j}\sin\frac{\Pi}{2}\right) + (-1)(\cos\Pi - j\sin\Pi) + (1)\left(\cos\frac{3\Pi}{2} - j\sin\frac{3\Pi}{2}\right)$$

$$2 + 3(-j) + (-1)(-1) + (1)(j)$$

$$2 - 3j + 1 + j$$

$$x[2] = \sum_{n=0}^{3} x\,[n]e^{-j2\Pi}\frac{(2)(n)}{4}$$

$$x[0]e^{-j2\Pi}\frac{(2)(0)}{4} + x[1]e^{-j2\Pi}\frac{(2)(1)}{4} + x[2]e^{-j2\Pi}\frac{(2)(2)}{4} + x[3]e^{-j2\Pi}\frac{(2)(3)}{4}$$

$$x[0] + x[1]e^{-j\Pi} + x[2]e^{-j2\Pi} + x[3]^{-j3\Pi}$$

$$2 + (3)(-1) + (-1)(1) + (1)(-1)$$

$$2 - 3 - 1 - 1$$

$$-3$$

$$x[3] = \sum_{n=0}^{3} x\,[n]e^{-j_{2\Pi}\frac{3n}{4}}$$

$$x[0]e^{-j2\Pi}\frac{(3)(0)}{4} + x[1]e^{-j2\Pi}\frac{(3)(1)}{4} + x[2]e^{-j2\Pi}\frac{(3)(2)}{4} x[3]e^{-j2\Pi}\frac{(3)(3)}{4}$$

$$x[0] + x[1]e^{-j\frac{\Pi}{2}} + x[2]e^{-j^{3\Pi}} + x[3]e^{-j\frac{9\Pi}{2}}$$

$$(2) + (3)(j) + (-1)(-1) + (1)(-j)$$

$$2 + 3j + 1 - j$$

$$3 + 2j$$

$$\boxed{\underline{x[k] = \{5{,}3 - 2j, 3{,}3 + 2j\}}}$$

DISCRETE FOURIER TRANSFORM (CONT.)

$$\begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix} = \begin{bmatrix} w_0^4 & w_0^4 & w_0^4 & w_0^4 \\ w_0^4 & w_1^4 & w_2^4 & w_3^4 \\ w_0^4 & w_2^4 & w_4^4 & w_6^4 \\ w_0^4 & w_3^4 & w_6^4 & w_9^4 \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix}$$

$w_n = e^{-j\frac{2\Pi n}{N}}$ $\quad$ $w_3 = e^{-j\frac{2\Pi(3)}{4}}$ $\quad$ $w_9 = e^{-j\frac{2\Pi(9)}{4}}$

$w_0 = e^{-j\frac{2\Pi(0)}{4}}$ $\quad$ $w_3 = e^{-j\frac{2\Pi}{2}}$ $\quad$ $w_9 = e^{-j\frac{9\Pi}{2}}$

$w_0 = 1$ $\quad$ $w_3 = j$ $\quad$ $w_9 = -j$

$w_1 = e^{-j\frac{2\Pi(1)}{4}}$ $\quad$ $w_4 = e^{-j\frac{2\Pi(4)}{4}}$

$w_1 = e^{-j\frac{\Pi}{2}}$ $\quad$ $w_4 = e^{-j2\Pi}$

$w_1 = -j$ $\quad$ $w_4 = 1$

$w_2 = e^{-j\frac{2\Pi(2)}{4}}$ $\quad$ $w_6 = e^{-j\frac{2\Pi(6)}{4}}$

$w_2 = e^{-j\Pi}$ $\quad$ $w_6 = e^{-j3\Pi}$

$w_2 = -1$ $\quad$ $w_6 = -1$

$$\begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ -1 \\ 1 \end{bmatrix}$$

$x[0] = (1)(2) + (1)(3) + (1)(-1) + (1)(1)$

$x[0] = 2 + 3 - 1 + 1 = 5$

$x[1] = (1)(2) + (j)(3) + (-1)(-1) + (j)(1)$

$x[1] = 2 + 3j + 1 + j = 3 - 2j$

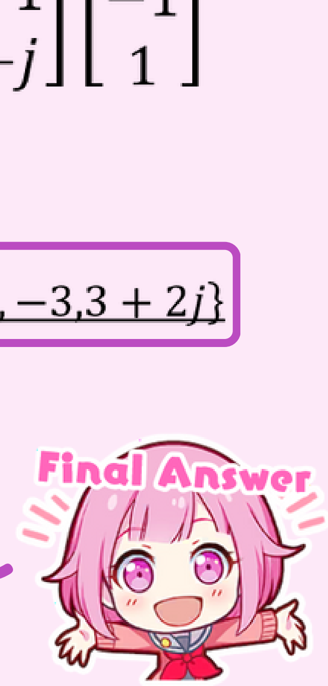
$x[2] = (1)(2) + (-1)(3) + (1)(-1) + (-1)(1)$

$x[2] = 2 - 3 - 1 - 1 = -3$

$x[3] = (1)(2) + (j)(3) + (-1)(-1) + (-j)(1)$

$x[3] = 2 + 3j + 1 - j = 3 + 2j$

$$\boxed{\underline{x[k] = \{5, 3 - 2j, -3, 3 + 2j\}}}$$



INVERSE DFT

$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} x\,[k]e^{j2\pi\frac{k(n)}{N}}$$

$$x[k] = [5{,}3 - 2j, -3{,}3 + 2j]$$

$$x[0] = \frac{1}{4}\sum_{k=0}^{3} x\,[k]e^{j2\pi\frac{k(0)}{4}}$$

$$x[0] = \frac{1}{4}\left[x[0]e^{j2\pi\frac{(0)(0)}{4}} + x[1]e^{j2\pi\frac{(1)(0)}{4}} + x[2]e^{j2\pi\frac{(2)(0)}{4}} + x[3]e^{j2\pi\frac{(3)(0)}{4}}\right]$$

$$x[0] = \frac{1}{4}[x[0] + x[1] + x[2] + x[3]]$$

$$x[0] = \frac{1}{4}[5 + (3 - 2j) + (-3) + (3 + 2j)]$$

$$x[0] = \frac{1}{4}[5 + (3 - 2j) + (-3) + (3 + 2j)]$$

$$x[0] = \frac{1}{4}[5 + 3] = \frac{1}{4}(8)\boxed{= 2}$$

$$x[1] = \frac{1}{4}\left[x[0]e^{j2\pi\frac{(0)(1)}{4}} + x[1]e^{j2\pi\frac{(1)(1)}{4}} + x[2]e^{j2\pi\frac{(2)(1)}{4}} + x[3]e^{j2\pi\frac{(3)(1)}{4}}\right]$$

$$x[1] = \frac{1}{4}\left[x[0] + x[1]e^{j\frac{\pi}{2}} + x[2]e^{j\pi} + x[3]e^{j\frac{3\pi}{2}}\right]$$

$$x[1] = \frac{1}{4}[5 + (3 - 2j)(j) + (-3)(-1) + (3 + 2j)(-j)]$$

$$x[1] = \frac{1}{4}[5 + (3j - 2j^2) + 3 -, (-3 - 2j^2)]$$

$$x[1] = \frac{1}{4}[8 - 4j^2]$$

$$x[1] = \frac{1}{4}[8 - 4(-1)] = \frac{1}{4}[8 + 4] = \frac{1}{4}(12)\boxed{= 3}$$

$$x[2] = \frac{1}{4}\left[x[0]e^{j2\pi\frac{(0)(2)}{4}} + x[1]e^{j2\pi\frac{(1)(2)}{4}} + x[2]e^{j2\pi\frac{(2)(2)}{4}} + x[3]e^{j2\pi\frac{(3)(2)}{4}}\right]$$

$$x[2] = \frac{1}{4}[x[0] + x[1]e^{j\pi} + x[2]e^{j2\pi} + x[3]e^{j3\pi}]$$

$$x[2] = \frac{1}{4}[5 + (3 - 2j)(-1) + (-3)(1) + (3 + 2j)(-1)]$$

$$x[2] = \frac{1}{4}[5 + (3 + 2j) + (-3) + (-3 - 2j)]$$

$$x[2] = \frac{1}{4}\ 5 + (-3) + (-3) + (-3)$$

$$x[2] = \frac{1}{4}[5 - 9]$$

$$x[2] = \frac{1}{4}[-4]\boxed{= -1}$$

INVERSE DFT
(CONT.)

$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} x\,[k]e^{j2\pi}\,\frac{k(n)}{N}$$

$$x[k] = [5{,}3 - 2j, -3{,}3 + 2j]$$

$$x[3] = \frac{1}{4}\left[x[0]e^{j2\pi}\,\frac{(0)(3)}{4} + x[1]e^{j2\pi}\,\frac{(1)(3)}{4} + x[2]e^{j2\pi}\,\frac{(2)(3)}{4} + x[3]e^{j2\pi}\,\frac{(3)(3)}{4}\right]$$

$$x[3] = \frac{1}{4}\left[x[0] + x[1]e^{\frac{3\pi}{2}} + x[2]e^{3\pi} + x[3]e^{j}\,\frac{9\pi}{2}\right]$$

$$x[3] = \frac{1}{4}[5 + (3 - 2j)(-j) + (-3)(-1) + (3 + 2j)(j)]$$

$$x[3] = \frac{1}{4}[5 + (-3 - 2j^2) + (-3) + (3j + 2j^2)]$$

$$x[3] = \frac{1}{4}[8 + 4j^2]$$

$$x[3] = \frac{1}{4}[8 + 4(-1)] = \frac{1}{4}[8 - 4] = \frac{4}{4}\boxed{= 1}$$

FOURIER
SERIES

- A Fourier series is an expansion of a periodic function f(x) in terms of an infinite sum of sines and cosines
- Fourier series make use of the orthogonality relationships of the sine and cosine functions
- The computation and study of Fourier series is known as harmonic analysis and is extremely useful to break up an arbitrary periodic function into a set of simple terms that can be plugged in, solved individually, and then recombined to obtain the solution to the original problem or an approximation to it to whatever accuracy is desired or practical

Joseph Fourier, in full Jean-Baptiste-Joseph, Baron Fourier, (born March 21, 1768, Auxerre, France—died May 16, 1830, Paris)

FOURIER SERIES
WITH PERIOD 2Π

(ii) Let $f(x)$ be periodic with period 2π and piecewise continuous in the interval $[-\pi, \pi]$. Then the Fourier series of $f(x)$ is given by

$$\boxed{f(x) = a_0 + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx)}$$

with coefficients

$$\left.\begin{aligned} a_0 &= \tfrac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx, \\ a_n &= \tfrac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx, \quad n = 1, 2, 3, \ldots \\ b_n &= \tfrac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx, \quad n = 1, 2, 3, \ldots \end{aligned}\right\}$$

EVEN & ODD FUNCTIONS

Fourier series takes on simpler forms for Even and Odd functions

A function is Even if f(x) = f(-x) for all x.
The graph of an even function is
symmetric about the y-axis.

A function is Odd if f(x) = -f(-x) for all x.
The graph of an even function is
skew- symmetric about the y-axis.

- Let f(x) be a function of period 2 such that f(x)= x/2 [0, 2π].
- Find the Fourier series of f(x) over [0, 2 pi]

$$p = 2\Pi$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$a_0 = \frac{1}{2\Pi} \int_0^{2\Pi} \frac{x}{2} dx$$

$$a_0 = \frac{1}{4\Pi} \int_0^{2\Pi} x\, dx$$

$$a_0 = \frac{1}{4\Pi} \left(\frac{x^2}{2} \right) \Big|_0^{2\Pi}$$

$$a_0 = \frac{1}{4\Pi} \left[\frac{(2\Pi)^2}{2} - \frac{(0)^2}{2} \right]$$

$$a_0 = \frac{1}{4\Pi} \left(\frac{4\Pi^2}{2} \right)$$

$$a_0 = \frac{\Pi}{2}$$

$$a_n = \frac{1}{\Pi} \int_0^{2\Pi} \left(\frac{x}{2} \right) \cos nx\, dx$$

$$a_n = \frac{1}{2\Pi} \int_0^{2\Pi} x \cos nx\, dx$$

$$\int u\, dv = uv - \int v\, du$$

$$u = x \quad dv = \cos nx\, dx$$
$$du = dx \quad v = \frac{1}{n} \sin nx$$

$$a_n = \frac{1}{2\Pi} \left\{ \left[\frac{x}{n} \sin nx \right] \Big|_0^{2\Pi} - \frac{1}{n} \int_0^{2\Pi} \sin\ nx \right\}$$

$$a_n = \frac{1}{2\Pi} \left\{ \left[\frac{1}{n} (2\Pi \sin 2\Pi n - 0 \sin(0)) \right] + \frac{1}{n^2} (\cos nx)|_0^{2\Pi} \right\}$$

$$a_n = \frac{1}{2\Pi} \left[\frac{1}{n^2} (\cos 2\Pi n - \cos n(0)) \right]$$

$$a_n = \frac{1}{2\Pi} \left[\frac{1}{n^2} (1 - 1) \right]$$

$$a_n = \frac{1}{2\Pi} \left[\frac{1}{n^2} (0) \right]$$

$$a_n = 0$$

$$b_n = \frac{1}{\Pi} \int_0^{2\Pi} \frac{x}{2} \sin nx\, dx$$

$$b_n = \frac{1}{2\Pi} \int_0^{2\Pi} x \sin x\, dx$$

$$u = x \quad dv = \sin nx\, dx$$
$$du = dx \quad v = -\frac{1}{n} \cos nx$$

$$b_n = \frac{1}{2\Pi} \left[\left(-\frac{x}{n} \cos nx \right) \Big|_0^{2\Pi} + \frac{1}{n} \int_0^{2\Pi} \cos nx\, dx \right]$$

$$b_n = \frac{1}{2\Pi} \left[\left(-\frac{x}{n} \cos nx \right) \Big|_0^{2\Pi} + \left(\frac{1}{n^2} \sin nx \right) \Big|_0^{2\Pi} \right]$$

$$b_n = \frac{1}{2\Pi} \left[-\frac{1}{\Pi} (2\Pi \cos 2\Pi n - 0 \cos n(0)) + \frac{1}{n^2} (\sin 2\Pi - \sin n(0)) \right]$$

$$b_n = \frac{1}{2\Pi} \left[-\frac{1}{n} (2\Pi) \right]$$

$$b_n = -\frac{1}{n}$$

$$\boxed{f(x) = \frac{\Pi}{2} - \frac{1}{n} \sum_{n=1}^{\infty} \sin\ nx}$$

$$f(x) = \begin{cases} k, & 0 < x < \pi \\ -k, & -\pi < x < 0 \end{cases}$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$a_0 = \frac{1}{2\Pi}\left[\int_0^{\Pi} k\,dx + \int_{-\Pi}^{0} -k\,dx\right]$$

$$a_0 = \frac{1}{2\Pi}[(kx)|_0^{\Pi} + (-kx)|_{\Pi}^{0}]$$

$$a_0 = [[k_{\Pi} - k(0)] - k(0) + k(\Pi)]$$

$$a_0 = \frac{1}{2\Pi}[k\Pi - k\Pi]$$

$$a_0 = 0$$

$$a_n = \frac{1}{\Pi}\left[\int_0^{\Pi} k\cos nx\,dx + \int_{-\Pi}^{0} -k\cos nx\,dx\right]$$

$$a_n = \frac{1}{\Pi}\left[\frac{k}{n}(\sin nx)|_0^{\Pi} - \frac{k}{n}(\sin nx)|_{-\Pi}^{0}\right]$$

$$a_n = \frac{1}{\Pi}\left[\frac{k}{n}(\sin n \quad \Pi \quad - \sin n \quad (0)) - \frac{k}{n}\left(\sin n(0)^{+\sin n}\Pi\right)\right]$$

$$a_n = \frac{1}{\Pi}\left[\frac{k}{n} - \frac{k}{n}\right]$$

$$a_n = \frac{1}{\Pi}[0]$$

$$a_n = 0$$

$$b_n = \frac{1}{\Pi}\left[\int_0^{\Pi} k\sin nx\,dx + \int_{-\Pi}^{0} -k\sin nx\,dx\right]$$

$$b_n = \frac{1}{\Pi}\left[-\frac{k}{n}(\cos nx)|_0^{\Pi} + \frac{k}{n}(\cos nx)|_{-\Pi}^{0}\right]$$

$$b_n = \frac{1}{\Pi}\left[-\frac{k}{n}(\cos n\Pi - \cos n(0)) + \frac{k}{n}(\cos n(0) - \cos n\Pi)\right]$$

$$b_n = -\frac{k}{\Pi n}(-\cos n\Pi + 1 + 1 - \cos n\Pi)$$

$$b_n = \frac{2k}{\Pi n}(1 - \cos n\Pi)$$

$$f(x) = \sum_{n=1}^{\infty} \frac{2k}{\Pi n}(1 - \cos n\Pi)\sin nx$$

$$f(x) = \frac{2k}{\Pi}\sum_{n=1}^{\infty} \frac{(1 - \cos n\Pi)}{r}\sin nx$$

$$f(x) = \begin{cases} 1, & -\frac{\pi}{2} < x < \frac{\pi}{2} \\ -1, & \frac{\pi}{2} < x < \frac{3\pi}{2} \end{cases}$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$a_0 = \frac{1}{2\Pi}\left[\int_{\frac{-\Pi}{2}}^{\frac{\Pi}{2}} 1\,dx + \int_{\frac{\Pi}{2}}^{\frac{3\Pi}{2}} -1\,dx\right]$$

$$a_0 = \frac{1}{2\Pi}\left[\int_{\frac{-\Pi}{2}}^{\frac{\Pi}{2}} x\,dx + \int_{\frac{\Pi}{2}}^{\frac{3\Pi}{2}} -x\,dx\right]$$

$$a_0 = \frac{1}{2\Pi}\left\{\left(\frac{\Pi}{2} - \left(-\frac{\Pi}{2}\right)\right) + \left(-\frac{\Pi}{2} + \frac{3\Pi}{2}\right)\right\}$$

$$a_0 = \frac{1}{2\Pi}\left[\frac{2\Pi}{2} + \left(-\frac{2\Pi}{2}\right)\right]$$

$$a_0 = \frac{1}{2\Pi}[0]$$

$$a_0 = 0$$

$$a_n = \frac{1}{\Pi}\left[\int_{\frac{-\Pi}{2}}^{\frac{\Pi}{2}} (1)\cos nx\,dx + \int_{\frac{\Pi}{2}}^{\frac{3\Pi}{2}} (-1)\cos nx\,dx\right]$$

$$a_n = \frac{1}{\Pi}\left[\frac{1}{\Pi}(\sin nx)\Big|_{\frac{-\Pi}{2}}^{\frac{\Pi}{2}} - \frac{1}{\Pi}(\sin nx)\Big|_{\frac{\Pi}{2}}^{\frac{3\Pi}{2}}\right]$$

$$a_n = \frac{1}{\Pi}\left[\frac{1}{\Pi}(1-(-1)) - \frac{1}{\Pi}(-1-1)\right]$$

$$a_n = \frac{1}{\Pi}\left[\frac{1}{\Pi}(2) - \frac{1}{\Pi}(-2)\right]$$

$$a_n = \frac{1}{\Pi}\left(\frac{4}{\Pi}\right)$$

$$a_n = \frac{4}{\Pi n}$$

$$b_n = \frac{1}{\pi}\left(\int_{\frac{-\Pi}{2}}^{\frac{\Pi}{2}} \sin nx\,dx + \int_{\frac{\Pi}{2}}^{\frac{3\Pi}{2}} -\sin nx\,dx\right)$$

$$b_n = \frac{1}{\Pi}\left[-\frac{1}{\Pi}(\cos nx)\Big|_{\frac{-\Pi}{2}}^{\frac{\Pi}{2}} + \frac{1}{\Pi}(\cos nx)\Big|_{\frac{\Pi}{2}}^{\frac{3\Pi}{2}}\right]$$

$$b_n = \frac{1}{\Pi}\left[-\frac{1}{\Pi}(0) + \frac{1}{\Pi}(0)\right]$$

$$b_n = \frac{1}{\Pi}[0]$$

$$b_n = 0$$

$$\boxed{\begin{aligned} f(x) &= a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \\ f(x) &= \frac{4}{\Pi n}\sum_{n=1}^{\infty} \cos nx \\ f(x) &= \frac{4}{\Pi}\cos x + \frac{2}{\Pi}\cos 2x + \frac{4x}{3\Pi}\cos 3x + \frac{1}{\Pi}\cos 4x \end{aligned}}$$

DISCRETE TIME
FOURIER TRANSFORM

$$x[e^{j\omega}] = \sum_{n=-\infty}^{\infty} x\,[n]e^{-j\omega n}$$

$$x[n] = \frac{1}{2\Pi} \int_{-\pi}^{\pi} x(e^{j\omega})(e^{j\omega n})\, d\omega$$

$$\delta^{n} = 1$$

$$\delta(n-1) = e^{-j\omega} \times 1$$

$$\delta(n-2) = e^{-j2\omega} \times 1$$

$$\delta(n+2) = e^{j2\omega} \times 1$$

Goal: $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$

$$x(n) = a^n u(n); a < 1$$

$$e^{j\omega}] = \sum_{n=-\infty}^{\infty} \delta[n]e^{-j\omega n(0)}$$

$$e^{j\omega}] = \sum_{n=-\infty}^{\infty} a^n u[n]e^{-j\omega n}$$

$$x[e^{j\omega}] = \sum_{n=0}^{\infty} a^n(1)e^{-j\omega n}$$

$$x[e^{j\omega}] = \sum_{n=0}^{\infty} (a\, e^{-j\omega})^n$$

$$\boxed{\frac{1}{1-\frac{e^{j\omega}}{ae^{-j\omega}}}}$$

$$x(n) = 2^n u(-n)$$

$$x[e^{j\omega}] = \sum_{n=-\infty}^{\infty} 2^n u[-n]e^{-j\omega n}$$

$$x[e^{j\omega}] = \sum_{n=-\infty}^{0} 2^n(1)e^{-j\omega n}$$

$$x[e^{j\omega}] = \sum_{n=0}^{\infty} 2^{-n}e^{j\omega n}$$

$$x[e^{j\omega}] = \sum_{n=0}^{\infty} (2^{-1}\, e^{j\omega})^n$$

$$x[e^{j\omega}] = \sum_{n=0}^{\infty} \left(\frac{e^{j\omega}}{2}\right)^n$$

$$\boxed{\frac{1}{1-\frac{e^{j\omega}}{2}}}$$

$$x(n) = \left(\frac{1}{3}\right)^n u(n-3)$$

$$x[e^{j\omega}] = \sum_{n=-\infty}^{\infty} \left(\frac{1}{3}\right)^n u[n-3]e^{-j\omega n}$$

$$x[e^{j\omega}] = \sum_{n=3}^{\infty} \left(\frac{1}{3}\right)^n (1)e^{-j\omega n}$$

$$m = n-3$$

$$n = m+3$$

$$x[e^{j\omega}] = \sum_{m=0}^{\infty} \left(\frac{1}{3}\right)^{m+3} e^{-j\omega(m+3)}$$

$$x[e^{j\omega}] = \sum_{m=0}^{\infty} \left(\frac{1}{3}\right)^3 \left(\frac{1}{3}\right)^m e^{-j3\omega}e^{-j\omega m}$$

$$\left(\frac{1}{27}\right)(e^{-j3\omega}) = \sum_{m=0}^{\infty} \left(\frac{e^{j\omega}}{3}\right)^m$$

$$\left(\frac{e^{-j3\omega}}{27}\right)\left(\frac{1}{1-\frac{e^{-j\omega}}{3}}\right) \qquad \boxed{\left(\frac{e^{-j3\omega}}{27\left(1-\frac{e^{-j\omega}}{3}\right)}\right)}$$

$$x(n) = \delta[n]$$

$$\boxed{= 1}$$

DISCRETE TIME
FOURIER TRANSFORM

1. $x(e^{j\omega}) = \frac{6-\frac{2}{3}e^{-j\omega}-\frac{1}{6}e^{-2j\omega}}{-\frac{1}{6}e^{-j2\omega}+\frac{1}{6}e^{-j\omega}+1}$

Find x(n)

$$(e^{j\omega})^2 = (t)^2$$

$$x(e^{j\omega}) = \frac{6-\frac{2}{3}t-\frac{1}{6}t^2}{-\frac{1}{6}t^2+\frac{1}{6}t+1}$$

$$x(e^{j\omega}) = \frac{-\frac{1}{6}t^2-\frac{2}{3}t+6}{-\frac{1}{6}t^2+\frac{1}{6}t+1} * \frac{-6}{-6}$$

$$x(e^{j\omega}) = \frac{t^2+4t-36}{t^2-t-6}$$

$$1+\frac{5t-30}{t^2-t-6}$$

$$1+\frac{5t-30}{(t-3)(t+2)}$$

$$= \frac{A}{t-3}+\frac{B}{t+2}$$

$$5t-30 = A(t+2)+B(t-3)$$

$$5 = A+B \rightarrow 15 = 3A+3B$$

$$-30 = 2A+2B \rightarrow -30 = 2A+3B$$

$$-15 = 5A$$

$$A = -3$$

$$B = 5-A$$

$$B = 5-(-3)$$

$$B = 8$$

$$= 1+\frac{-3}{t-3}+\frac{8}{t+2}$$

$$= 1+\frac{-3}{-(3-t)}+\frac{8}{2-(-t)}$$

$$= 1+\frac{-3}{-3-t}*\frac{\frac{1}{3}}{\frac{1}{3}}+\frac{8}{2-(-t)}*\frac{\frac{1}{2}}{\frac{1}{2}}$$

$$= 1+\frac{1}{1-\frac{t}{3}}+4\left(\frac{1}{1-\left(-\frac{t}{2}\right)}\right)$$

$$= 1+\frac{1}{1-e\frac{-j\omega}{3}}+4\left(\frac{1}{1-\frac{-e^{-j\omega}}{2}}\right)$$

$$\boxed{\delta[n]+\left(\frac{1}{3}\right)^n u(n)+2/\left(-\frac{1}{2}\right)^n u[n]}$$

2. $x(e^{j\omega}) = \frac{3-\left(\frac{1}{4}\right)e^{-j\omega}}{-\frac{1}{16}e^{-2j\omega}+1}$

$find\ (x)$

$$x(e^{j\omega}) = \frac{3-\frac{1}{4}t}{-\frac{1}{16}t^2+1}$$

$$= \frac{-\frac{1}{4}t+3}{-\frac{1}{16}t^2+1}*\frac{16}{16}$$

$$= \frac{4t-48}{t^2-16}$$

$$= \frac{A}{t-4}+\frac{B}{t+4}$$

$$4t-48 = A(t+4)+B(t-4)$$

$$4 = A+B \rightarrow 16 = 4A+4B$$

$$-48 = 4A-4B \rightarrow -48 = 4A-4B$$

$$-32 = 8A$$

$$A = -4$$

$$B = 4-(4)$$

$$B = 8$$

$$= \frac{-4}{t-4}+\frac{8}{t+4}$$

$$= \frac{-4}{-(4-t)}*\frac{\frac{1}{4}}{\frac{1}{4}}+\frac{8}{4-(-t)}*\frac{\frac{1}{4}}{\frac{1}{4}}$$

$$= \frac{1}{1-\frac{t}{4}}+2\left(\frac{1}{1-\left(\frac{-t}{4}\right)}\right)$$

$$\boxed{= \left(\frac{1}{4}\right)^n u[n]+2\left(\frac{1}{4}\right)^n u[n]}$$

3. $x[e^{j\omega}] = e^{j\omega}$

$\delta[n^{+1}]$

$e^{j\omega} = \cos\omega + j\sin\omega$

$e^{-j\omega} = \cos\omega - j\sin\omega$

$$\frac{e^{j\omega}+e^{-j\omega}}{2} = \frac{2\cos\omega}{2}$$

$$\cos\omega = \frac{e^{j\omega}+e^{-j\omega}}{2}$$

$$\boxed{= \cos 2\omega = \frac{e^{j2\omega}+e^{-j2\omega}}{2}}$$

4. $x(e^{j\omega}) = \cos^2\omega$

$$x(e^{j\omega}) = \frac{(1+\cos 2\omega)}{2}$$

$$x(e^{j\omega}) = \frac{1}{2}+\frac{1}{2}\cos 2\omega$$

$$= \frac{1}{2}+\frac{1}{4}\left(e^{j2\omega}+e^{-j2\omega}\right)$$

$$\boxed{= \frac{1}{2}\delta[n]+\frac{1}{4}[\delta[n+2]+\delta[n-2]]}$$

CONVOLUTION

Perform the following on the sequence:

$$x(n) = \{1,2,3,1\}$$

$$h(n) = \{1,1,1\}$$

a. Linear Convolution

$$\begin{array}{r} 1\ 2\ 3\ 1 \\ \underline{\ 1\ 1\ 1} \\ 1\ 2\ 3\ 1 \\ 1\ 2\ 3\ 1\ x \\ \underline{1\ 2\ 3\ 1\ x\ x} \\ 1\ 3\ 6\ 6\ 4\ 1 \end{array}$$

$$\boxed{y[n] = \{1,3,6,6,4,1\}}$$

b. Circular Convolution

$$\begin{bmatrix} 1 & 1 & 3 & 2 \\ 2 & 1 & 1 & 3 \\ 3 & 2 & 1 & 1 \\ 1 & 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \\ 6 \\ 6 \end{bmatrix}$$

$$(1)(1) + (1)(1) + (3)(1) + (2)(0) = 5$$

$$(2)(1) + (1)(1) + (1)(1) + (3)(0) = 4$$

$$(3)(1) + (2)(1) + (1)(1) + (1)(0) = 6$$

$$(1)(1) + (3)(1) + (1)(1) + (1)(0) = 6$$

$$\boxed{y[n] = \{5,4,6,6\}}$$

c. Linear Convolutional using Circular Convolutional

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 3 & 2 \\ 2 & 1 & 0 & 0 & 1 & 3 \\ 3 & 2 & 1 & 0 & 0 & 1 \\ 1 & 3 & 2 & 1 & 0 & 0 \\ 0 & 1 & 3 & 2 & 1 & 0 \\ 0 & 0 & 1 & 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 6 \\ 6 \\ 4 \\ 1 \end{bmatrix}$$

Determine the linear convolution of the following system signals

$$x(n) = \delta(n) + 3\delta(n-1) + 2\delta(n-1) - \delta(n-3) + \delta(n-4)$$

$$h(n) = 2\delta(n) - \delta(n-1)$$

$$x(n) = \{1,3,2,-1,1\}$$

$$h(n) = \{2,-1\}$$

$$\begin{array}{r} 1\ 3\ 2-1\ 1 \\ \underline{\quad\quad 2-1} \\ -1-3-2\ 1-1 \\ \underline{2\ 6\ 4-2\ 2\ x} \\ 2\ 5-1-4\ 3-1 \end{array}$$

$$\boxed{y(n) = \{2,5,-1,-4,3,-1\}}$$